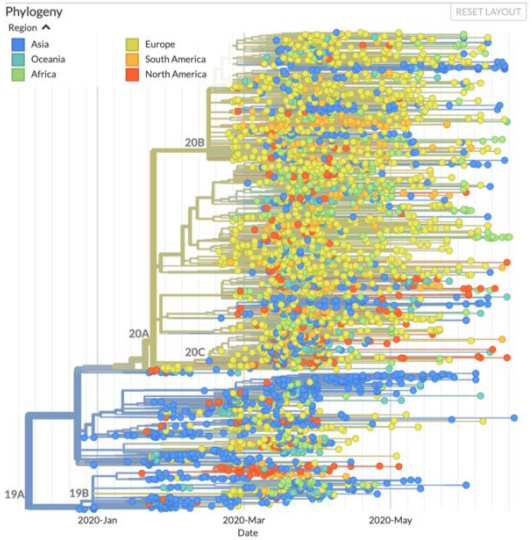
Epidemiological Applications of Phylogenetics

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Introduction

This session will cover:



- Introducing the relationship between phylogenetics and infectious disease epidemiology
- Phylogenetics to estimate parameters for how an infection spreads
- Phylogenetics to reconstruct where a pathogen was in the past
- Phylogenetics to reconstruct who infected whom
- Future directions and challenges

Phylogenetics and epidemiology of infectious disease

- Epidemiological applications of phylogenetics: the use of pathogen sequences to understand infectious disease transmission and dynamics.
- This sub-field is sometimes called *genomic epidemiology*
- “Transmission and dynamics” can mean:
  - pathogen origin (what are the likely sources),
  - time or origin, and for how long has it been undetected
  - infection population dynamics and epidemiological parameters
  - geographic movements of the pathogen
  - person-to-person transmission - who infected or did not infect whom
- Sequence data offer a high-resolution view of how pathogens spread, but the resolution is not perfect.

Whole Genome Sequences

Suppose for sampled pathogens we have a sequence of the whole genome.

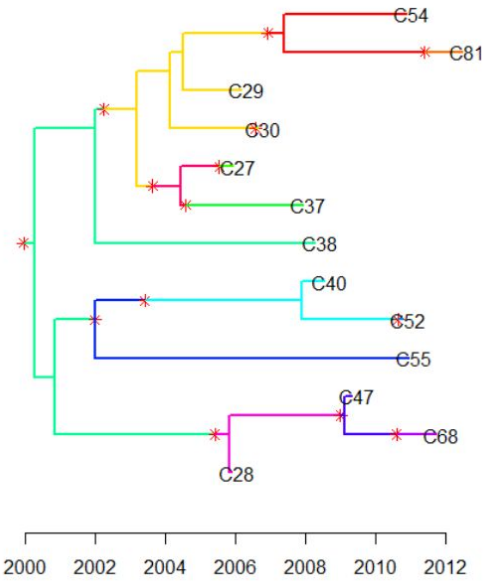
We can use this to estimate how closely related people's infections are.

Challenge: Use this information to make inferences about transmission

The density of sampling shapes what kinds of questions we can answer with genomic data.

Sequence 1: A A C T A G G T A G

Sequence 2: A G T T A G G C G G

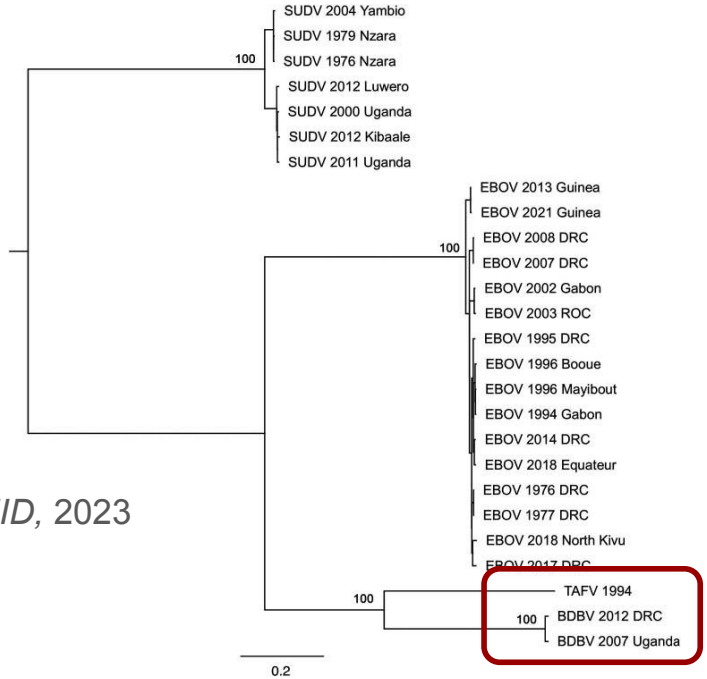


Questions phylogenetics helps answer nearly immediately

- What type or sub-type is this pathogen?
- How long has it been circulating (in humans)?
- What are its nearest known relatives?
- Is it likely to have been a new introduction from an animal population (a *zoonotic* introduction)?
- Or is it likely to be the result of (undetected) ongoing transmissing within humans?

EXAMPLE: Ebola outbreak ongoing in the Democratic Republic of Congo

Bundibugyo Ebola virus outbreak



[Judson et al](), *JID*, 2023

A1: The 2025 DRC outbreak is due to the **Bundibugyo strain**

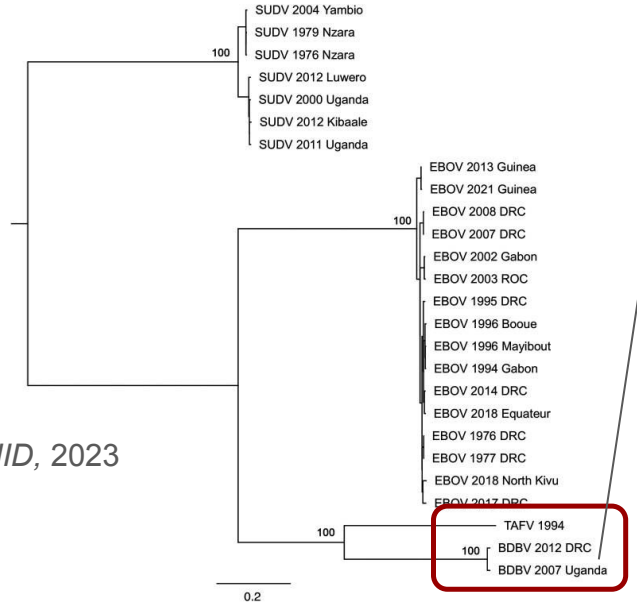
Ebola: Caused by viruses in the *Ortheobolavirus* genus

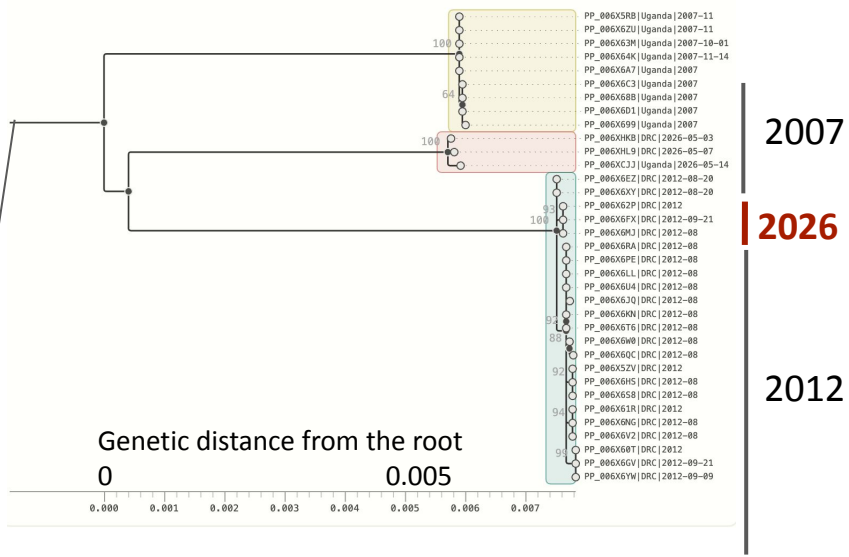
- Of the 6 viruses in the genus, only 4 cause Ebola in humans:
  - Bundibugyo virus
  - Sudan virus
  - Taï Forest virus
  - Ebola virus (formerly Zaire Ebola virus)
- Close relative of Marburg viruses
- Single-stranded negative-sense RNA virus
- Mostly does not spread human-to-human but persists in animal reservoirs (not entirely known which ones)
- Causes severe hemorrhagic fever, death (50% CFR)
- Spreads via bodily fluids, close contact, including at funerals (from the deceased)

Placing our outbreak in the phylogeny tells us about its origin

Ebola virus overall phylogeny; BDBV zoom in

First 3 sequences
virological.org May 18 2026
Shown with 34 other BDBV sequences





Judson et al, *JID*, 2023

A2: The 2026 cases are likely due to a **new introduction** from an animal reservoir population

A3: Nearest relatives: BDBV outbreaks; Uganda 2007 and DRC 2012

Timed phylogenetic trees

A timed phylogenetic tree has branches in units of calendar time.

These are constructed by combining phylogenetic reconstruction with the assumption of a *molecular clock:* at what rate do substitutions occur, in time? $d = \mu t$

The clock rate μ can best estimated with regression, using sequences gathered over time

Given a clock rate estimate, tools like Least Squares Dating can optimize the tree's branch lengths, giving the best fit to $d = \mu t$, scaling lengths to be in calendar time

This is also done simultaneously in Bayesian phylogenetics (which we will come to)

$$P(\text{tree, times, } \mu \mid \text{sequences}) \propto \mathbf{P(sequences \mid tree, w, \mu)\ P(tree \mid times, \theta)}\ P(\mu)\ P(w)\ P(\theta)$$

phylo likelihood **connection to epidemiology** clock

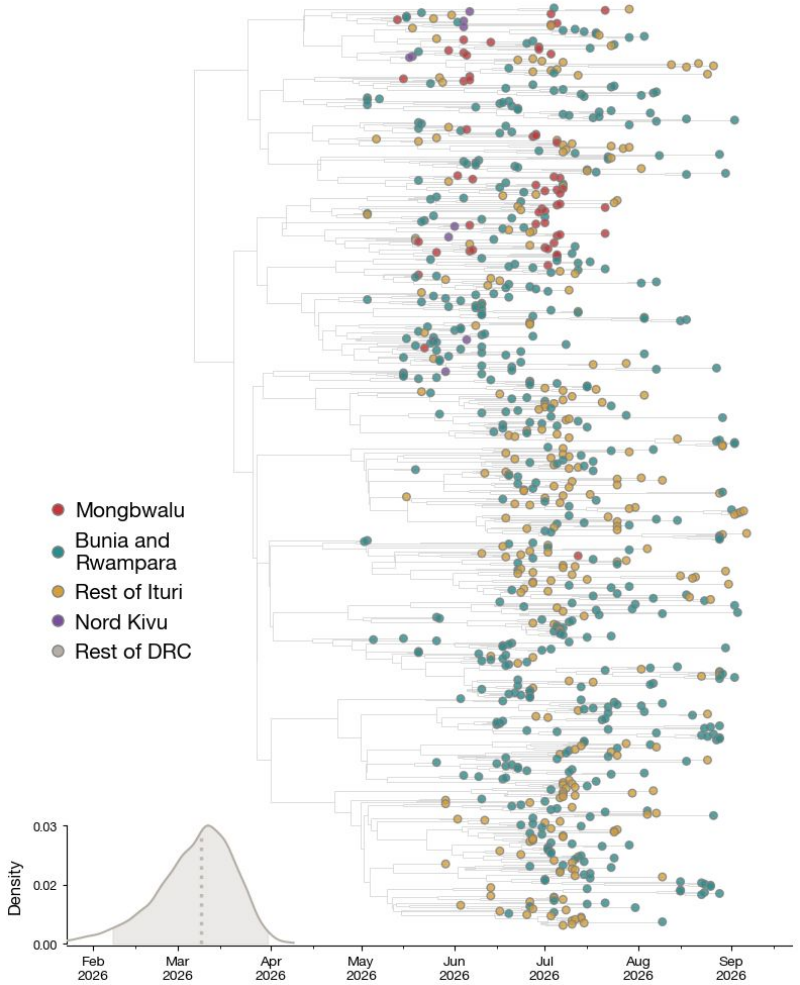
Example: DRC Ebola outbreak as of

Right: Timed phylogenetic tree.

Tips are coloured by location.

Q4: When did this outbreak likely begin?

A4: The tree indicates the likely time of origin: early March, 2026; there is some uncertainty.



Source: virological.org [Post on October 12, 2026]

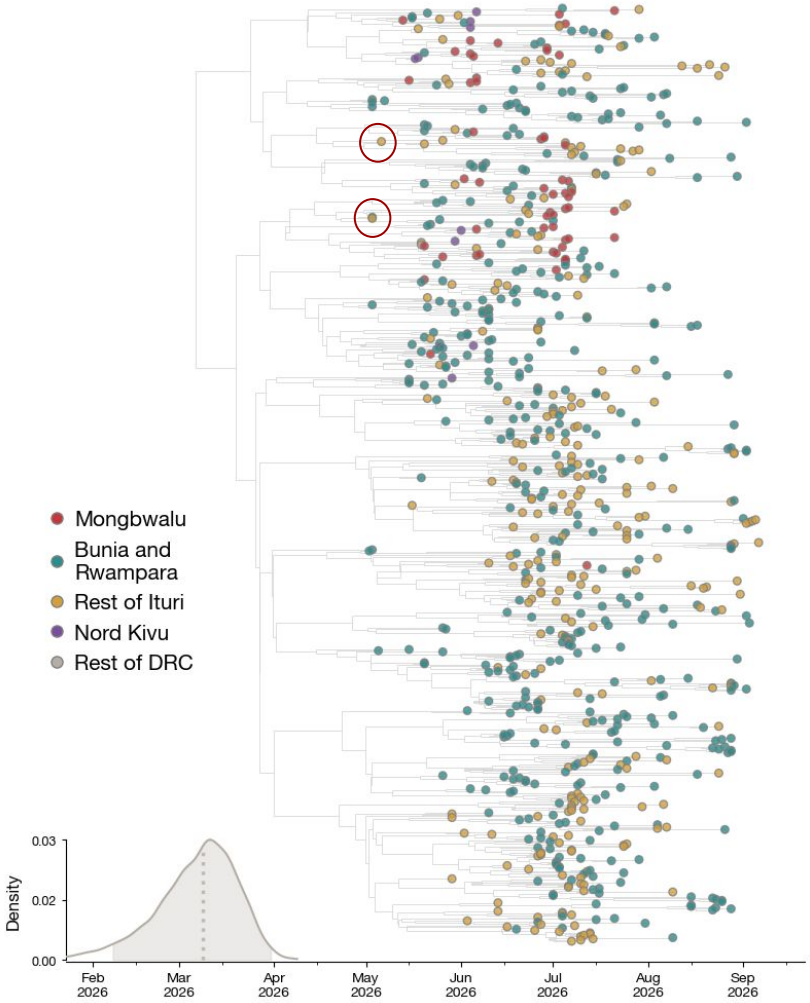
Quiz

Consider the two circled cases. They were found in the same region at the same time.

How much time elapsed between their most recen common ancestor and the time they were detecte

Do you think that their virus genomes are particularly closely related (within the outbreak?)

How many transmissions likely separate them?



Beyond the immediate questions: phylodynamics

The term *phylodynamics* usually refers to estimating one or more parameter describing the pathogen's population dynamics, using sequence data. Usually (but not always), sequences are collected over time, and we use a Bayesian inference framework with a posterior:

$$P(\text{tree, times, } \mu, \text{ w, } \theta \mid \text{seqs}) \propto \mathbf{P(seqs \mid tree, w, \mu)\ P(tree \mid \theta)}\ P(\mu)\ P(w)\ P(\theta)$$

phylo likelihood **connection to epidemiology** clock

It's worth talking about:

1. Bayesian phylogenetics

2. where ***P(tree | θ)*** comes from: it's the main link to transmission dynamics.

Aside: Bayesian phylogenetics primer

$P(tree, times, \mu, w, \theta | seqs) \propto$ ***P(seqs | tree, w, μ) P(tree | θ)*** $P(\mu)\, P(w)\, P(\theta)$

- ***P(sequences | tree, w, μ)****:* computed by Felsenstein's pruning under a molecular evolution model (HKY, JC69, etc) with molecular evolution model *w* and branch lengths ***μ t.***
- ***P(tree | θ):*** the tree prior (this links to the epidemiology). Coalescent or birth-death model gives the probability of the tree shape and node times given the epidemic process
- $P(\mu)\, P(w)\, P(\theta)$ priors on the clock rate, evolutionary model parameters, tree prior parameters.

This posterior is high-dimensional with discrete and continuous elements (tree topology + all node times + μ + substitution parameters + tree prior parameters) — no closed form

Explored by **MCMC**: sample trees and parameters proportional to their posterior probability. Tools: BEAST2, BEASTX, MrBayes.

Output: a posterior sample of trees → uncertainty in topology, dates, and parameters is propagated through to everything downstream

Phylodynamics: infer epidemiological parameters

The field of *phylodynamics* links a model with some parameters to the likelihood of a phylogenetic tree.

Then, either

- maximize the likelihood to find the best parameters, given a fixed phylogenetic tree created with data
- use that likelihood as a prior in the Bayesian decomposition to sample θ

$$P(\text{tree, times, } \mu, \text{ w, } \theta \mid \text{seqs}) \propto \mathbf{L(sequences \mid tree, w, \mu)}\ \mathbf{L(tree \mid \theta)}\ P(\mu)\ P(w)\ P(\theta)$$

The Bayesian route is more common, partly for historical reasons, and partly because at outbreak time scales, there can be a lot of uncertainty in the phylogeny.

The main scientific ingredient: *P(tree* | θ*)*

This is either the likelihood, if the tree is fixed, or the "tree prior", in Bayesian methods.

There are two major bodies of work developing P(tree | times, θ):

Coalescent models: condition on the sample size *n* (which should be << the population size *N*). Use an effective population size, *Ne* or *Ne(t)*, with the generation time $\boldsymbol{\tau}$, to write P(tree | times, θ).

Connects to population genetics, Wright-Fisher models.

Birth-death-sampling models: the tree is generated by a birth-death process with birth rate λ, death rate μ and sampling rate ψ (and sometimes other things)

Differences between birth-death and coalescent models

Coalescent: The likelihood is *L*(Tree | sampling times, *Ne(t)*)

The trajectory of Ne(t) does not affect how many samples there are, just the rates at which lineages coalesce.

Birth-death: The likelihood is *L*(Tree, sampling times | birth, death, sampling rate)

The birth, death and sampling rates *do* affect how many samples there are. The more cases you have, the more you are likely to sample.

Suppose you have lots of short branches and closely-related sequences:

- Coalescent: low Ne, this is nothing to worry about.
- Birth-death: this shows fast transmission, worry about this.

The tree likelihood for a birth-death-sampling model: estimating R from sequence data. R: mean number of new infections per infection.

Consider the stochastic process, beginning with one individual:

- Individuals give rise to new individuals at a birth rate λ
- Individuals are removed at a rate μ (recovery or death)
- Individuals are sampled at rate ψ (and this also removes them)

The process proceeds forward in time, and generates a tree.

In this tree, branching events correspond to transmission events. But not all transmission events are in the tree.

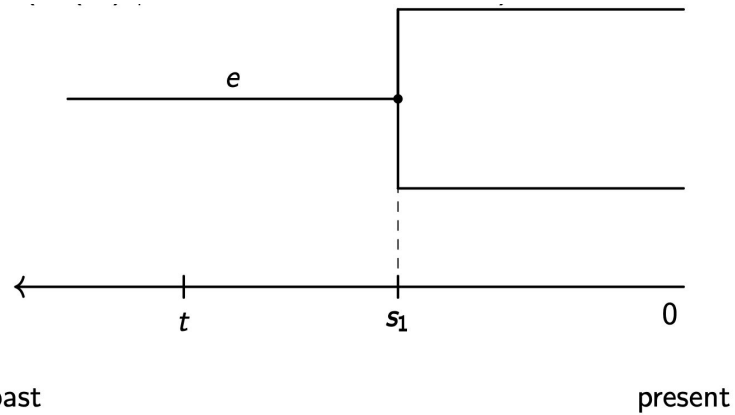
In this process, the *basic reproduction number* is $\lambda/(\mu + \psi)$.

We will build two key quantities: *g* and *p0*

$\mathcal{T}_e(s_1)$: the sub-tree descending from edge e at time s_1

$$p_0(t) = \mathcal{P}\left(\begin{array}{c} \text{a lineage has no sampled} \\ \text{or extant descendants} \end{array} \middle| \text{ the lineage is alive at time } t \right)$$

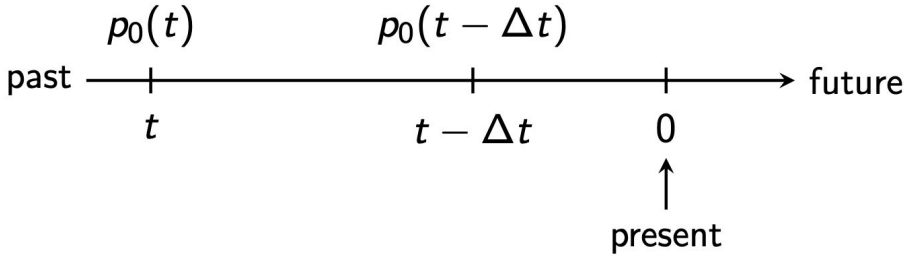
$$g_e(t) = \mathcal{P}(\mathcal{T}_e(s_1) \mid \text{lineage } e \text{ alive at time } t)$$



Differential equation approach

Because the birth-death process is Markovian, we can condition on what happens in the next small interval Δt and take the limit. This is a standard backward equation for a continuous-time Markov chain

Time convention: the present is time *t=0.*
Time increases into the past



Derivation of $p_0(t)$

We use the total law of probability to relate $p_0(t)$ to $p_0(t-\Delta t)$:

$$\underbrace{p_0(t)}_{P(\text{dead by end}|\text{alive at } t)} = \underbrace{\mu\Delta t}_{P(\text{death})} \cdot \underbrace{1}_{P(\text{dead by end}|\text{just died})} + \underbrace{\lambda\Delta t}_{P(\text{birth})} \cdot \left(\underbrace{p_0(t-\Delta t)}_{P(\text{dead by end}|\text{still alive})} \right)^2 + \underbrace{(1-(\mu+\lambda+\psi)\Delta t)}_{P(\text{nothing happens})} \cdot \underbrace{p_0(t-\Delta t)}_{P(\text{dead by end}|\text{still alive})}$$

$$p_0(t) = \mu\Delta t \times 1 + \lambda\Delta t p_0(t-\Delta t)^2 + (1-(\mu+\lambda+\psi)\Delta t)p_0(t-\Delta t),$$

$$\Longrightarrow$$

$$p_0(t) - p_0(t-\Delta t) = \left(\mu + \lambda p_0(t-\Delta t)^2 - (\mu+\lambda+\psi)p_0(t-\Delta t)\right)\Delta t,$$

$$\Longrightarrow$$

$$\frac{p_0(t) - p_0(t-\Delta t)}{\Delta t} = \left(\mu + \lambda p_0(t-\Delta t)^2 - (\mu+\lambda+\psi)p_0(t-\Delta t)\right).$$

This gives a differential equation for $p_0(t)$, the probability of having no descendants at the present time if alive at time t.

$$\frac{dp_0(t)}{dt} = \mu + \lambda p_0(t)^2 - (\lambda + \mu + \psi)p_0(t)^2$$

For a tip sampled at time $s > 0$, the initial condition for p_0 is

$$p_0(s) = 1$$

because sampling means being removed. There can be no descendant *at the present time* $t = 0$.

The DE for $p_0(t)$ does not depend on the tree, just the parameters. It can be solved explicitly.

$$p_0(t) = \frac{\lambda + \mu + \psi + c_1 \frac{e^{-c_1 t}(1-c_2)-(1+c_2)}{e^{-c_1 t}(1-c_2)+(1+c_2)}}{2\lambda}$$

We do the same for the quantity g

$$g_e(t) = \mathcal{P}(\mathcal{T}_e(s_1) \mid \text{lineage } e \text{ alive at time } t)$$

$$g_e(t+\Delta t) = (1-(\lambda+\mu+\psi)\Delta t - O(\Delta t^2))g_e(t) + \lambda \Delta t\, 2p_0(t)g_e(t) + O(\Delta t^2)$$

The first term: nothing happens.

The second term: There is a branching event, one side dies off ($p_0(t)$) and the other side produces the tree $\mathcal{T}_e(s_1)$. There are two ways this can happen (two descendants).

Taking $g_e(t+\Delta t) - g_e(t))$ and taking the limit as $\Delta t \to 0$ gives the differential equation for $g_e(t)$:

$$\frac{dg_e(t)}{dt} = -(\lambda+\mu+\psi)g_e(t) + 2\lambda p_0(t)g_e(t).$$

Initial conditions for g

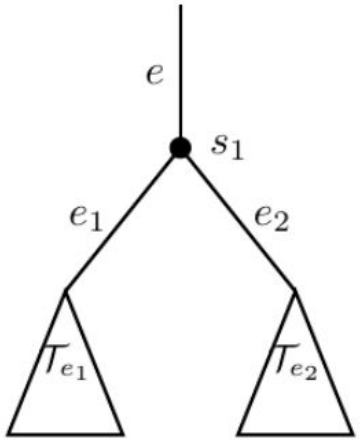
We still need initial conditions for the equations for $g_e(t)$.

Edge e ends either in a sampling event or in a branching event. Suppose e ends at time s_1. Set

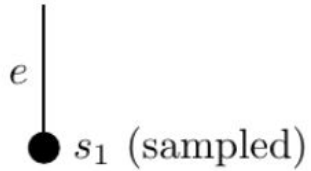
$$g_e(s_1) = \begin{cases} \lambda g_{e_1}(s_1) g_{e_2}(s_1) & \text{if } e \text{ has two descendant edges } e_1, e_2, \\ \psi & \text{if } e \text{ has no descendant edges.} \end{cases}$$

Up to this point $g_e(t)$ has been defined as a probability, but because sampling times are observed continuous random variables, the initialisation $g_e(s) = \psi$ makes g_e a density with respect to those times — so from here we treat it as a likelihood contribution rather than a probability. The ODE and its structure are unchanged; only the interpretation shifts at the tips.

Two cases for the initial conditions for *g*



$$g_e(s_1) = \lambda\, g_{e_1}(s_1)\, g_{e_2}(s_1)$$



$$g_e(s_1) = \psi$$

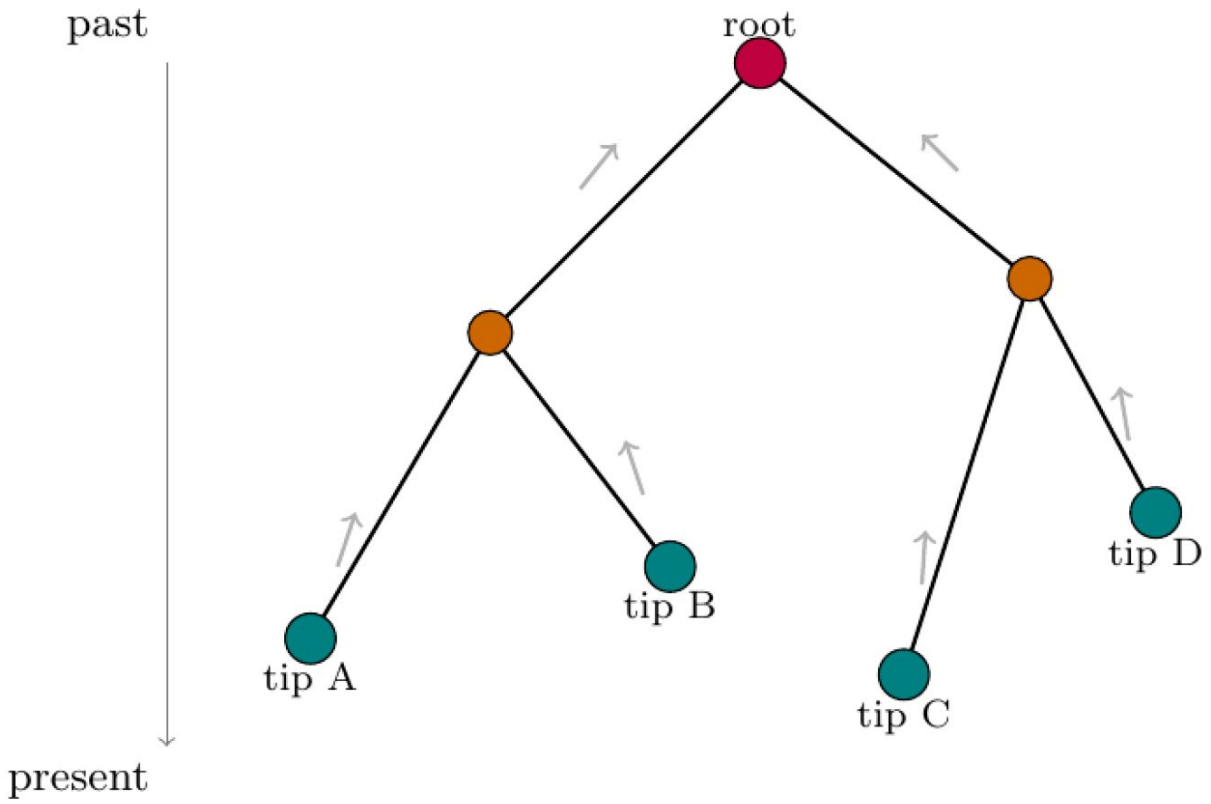
Stadler (2010) give the explicit solution to the ODE for *g*.

Building the tree likelihood for this birth-death-sampling model

- Solve the ODE for $p_0(t)$.
- Beginning at the tips, initialize $g_e(t)$ and solve the ODE for g_e, using the $p_0(t)$ solution, along the edges. You then obtain g_e for all terminal branches, at the times they originate.
- Continue working up the tree, setting $g_e(s_1) = \lambda g_{e_1}(s_1) g_{e_2}(s_1)$ at each branching event (joining the g solutions for edges e_1 and e_2 to start the equation for g on edge e
- When you reach the root of the tree, you have the tree likelihood!

$$L(\mathcal{T}|\lambda, \mu, \psi) = g_r(t_0)$$

where r is the root and t_0 is its time.



Step 1: solve $p_0(t)$ everywhere on tree

Step 2: tips $g_e(s) = \psi$ solve ODE upward

Step 3: at each node $g_e(s_1) = \lambda\, g_{e_1}(s_1) g_{e_2}(s_1)$ solve ODE upward

Step 4: root $\Rightarrow$ likelihood!

Now that you have $L(T|\theta)$ ..

For a fixed phylogenetic tree $\mathcal{T}$ you could find the parameters λ, μ and ψ for which the likelihood $g_r(t_0)$ is highest.

But $\mathcal{T}$ is not known completely. And there are some identifiability issues: the three parameters are not all identifiable.

The field uses Bayesian inference to jointly sample $\mathcal{T}$ with λ, μ and ψ. We can use prior knowledge, for example we probably know how long an infection lasts. This informs μ.

BEAST X, BEAST2: Propose a tree. Compute its likelihood. Compute the priors. Combine. Accept or reject. Continue. Result: a sample of trees and birth-death-sampling parameters.

Epidemiological parameters from phylogenetics

There are many models that can give a likelihood $L(T \mid \Theta)$.

Birth-death sampling variations include:

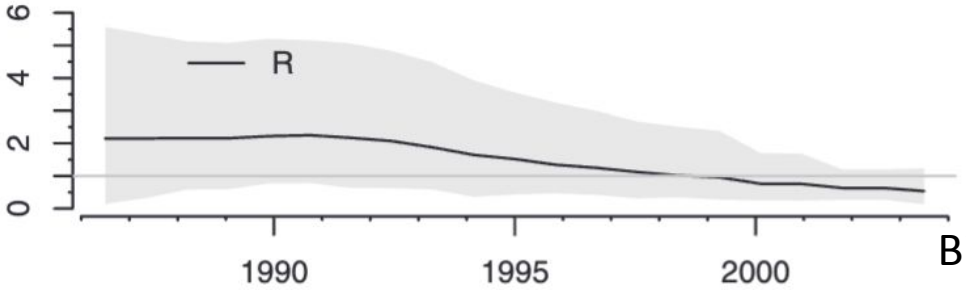
- “Sampled ancestors” (sampling does not remove the lineage - e.g. fossils)
- Multiple types, with different birth and death rates - like variants of concern
- Variable birth and death rates -- allows inference of reproduction number through time.

Coalescent variations include:

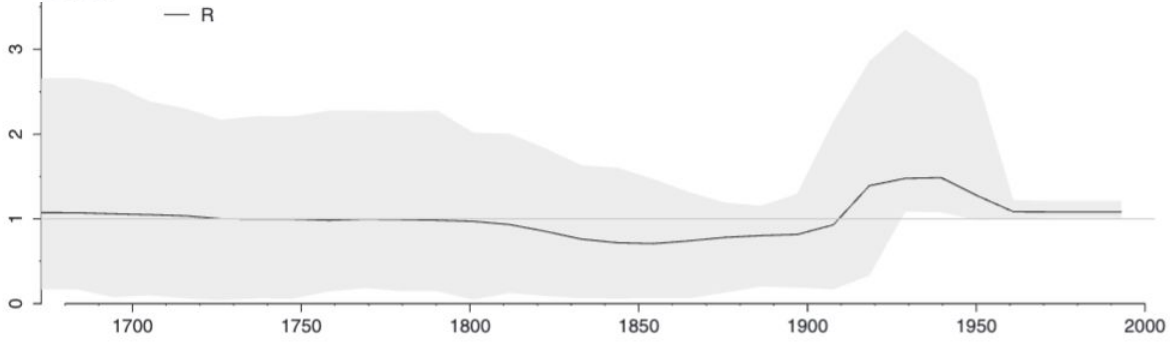
- Varying the effective population size through time, *Ne(t).*
- Epidemiological coalescent: susceptible-infectious-recovered (SIR) models, SEIR, etc
- Both incidence (new cases) *and* prevalence (*N*) contribute to coalescent rates
- “Structured populations” - lineages more likely to coalesce with others in the same “deme” (sub-population, such as a geographic region)

Application: R(t) in HIV and Hepatitis C (Stadler 2012)

HIV-1 subtype B in the United Kingdom; 62 sequences 1999-2003. R declines consistent with the development of multi-drug therapy.



HCV in Egypt: 63 sequences. Suspected cause of increase: contaminated injections against bilharziosis. Birth-death model consistent with this timing.



Short quiz

What is the main purpose of phylodynamics?

What are the two main sets of models used to set up tree likelihoods for phylodynamics?

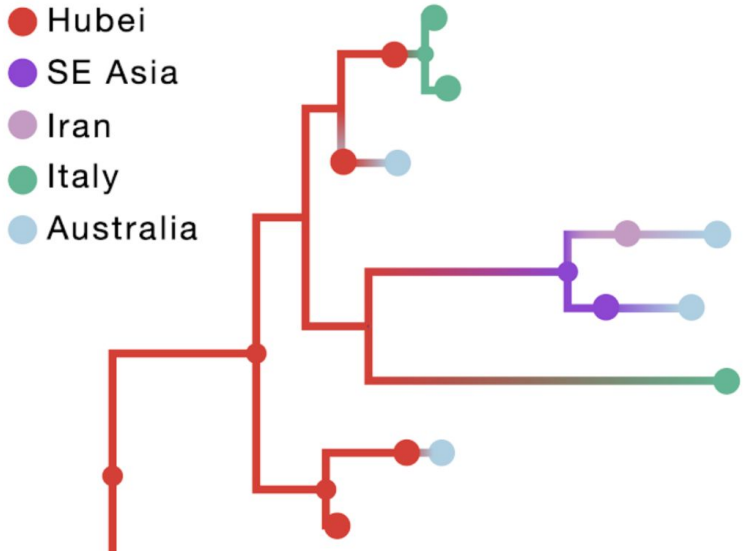
What is the most common inference approach?

Phylogeography

Phylogeography: estimating where pathogens were in the past.

“Where”: can be geographic location, or even host species.

Location can be discrete or continuous. Here: discrete state only.

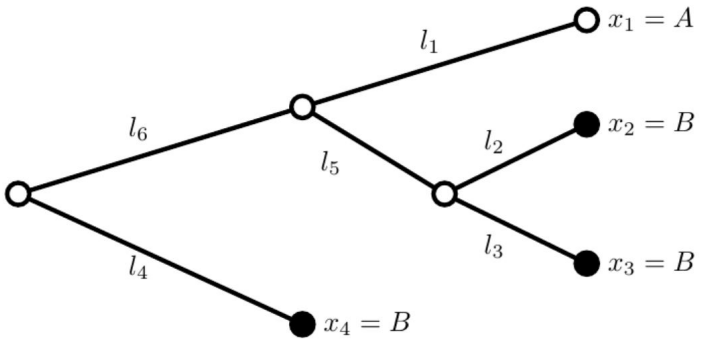


A two-location example illustrating the question

Here is a tree with 4 tips: 1, 2, 3, 4.

There are two locations: A (white) and B (black).

The question for discrete trait phylogeography is: where was the pathogen at the times of the internal nodes? (Phylogeography can also ask: where was the pathogen at any arbitrary point on an edge. But we will talk about only the internal nodes)



Key assumption: location can be modelled as evolving on the tree, just like a character in the DNA or RNA sequence.

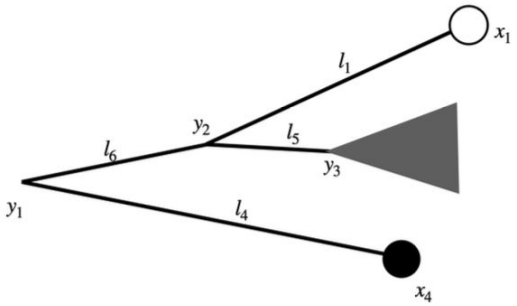
Location as a Markov process on the tree (like RNA or DNA)

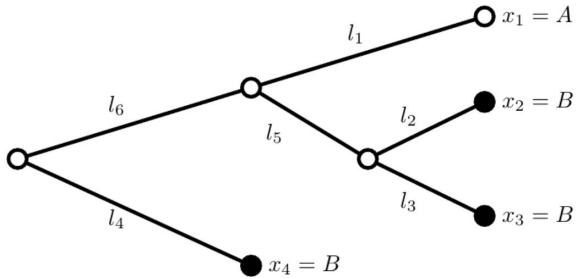
- The location of the pathogen at any point on the tree evolves as a continuous-time Markov chain with state space = set of locations $\{1, 2, \ldots, k\}$
- The rate matrix Q has entries q_{ij} = rate of moving from location i to location j, with $q_{ii} = -\sum_{j \neq i} q_{ij}$
- The transition probability matrix over a branch of length ℓ is $P(l) = e^{Ql}$, so $P_{ij}(\ell)$ = probability of being in location j at the end of a branch of length ℓ, given starting in location i
- Q is estimated from the data. The elements q_{ij} are the instantaneous rates of transition between states i and j. (Usually assumed to be symmetric).
- The likelihood is computed by Felsenstein pruning, exactly as for nucleotide substitution. Tip states are known. At each internal node, we sum over all possible ancestral locations
- The stationary distribution π of Q gives the long-run probability of being in each location, and we use that for the probability for the state of the root.

Recall the Felsenstein pruning algorithm

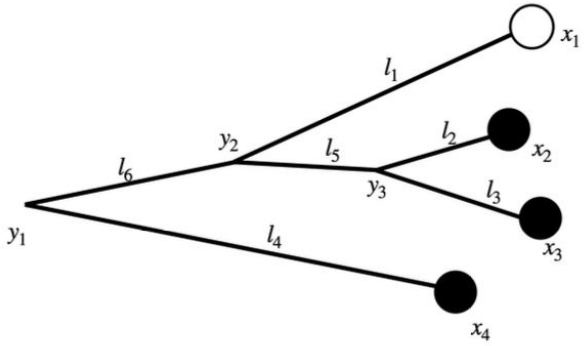
The notation $L_{y_k}(z)$ refers to the likelihood of the tree descending from node y_k, given that y_k has state z.

So we have as many terms $L_{y_3}(z)$ as there are possible values, z, for the node y_3





Recall Felsenstein's pruning (2)



$$L_{y_3}(z) = p(x_2 \mid y_3 = z, l_2)\, p(x_3 \mid y_3 = z, l_3)$$

$$L_{y_2}(z) = p(x_1 \mid y_2 = z, l_1) \left(\sum_w p(y_3 = w \mid y_2 = z, l_5)\, L_{y_3}(w) \right)$$

$$L_{y_1}(z) = p(x_4 \mid y_1 = z, l_4) \left(\sum_u p(y_2 = u \mid y_1 = z, l_6)\, L_{y_2}(u) \right)$$

Finally, the likelihood for the whole tree is the combination of likelihoods of the tree given that the root is in state z:

$$L_{y_1} = \sum_{y_1} \pi(z) L_{y_1}(z)$$

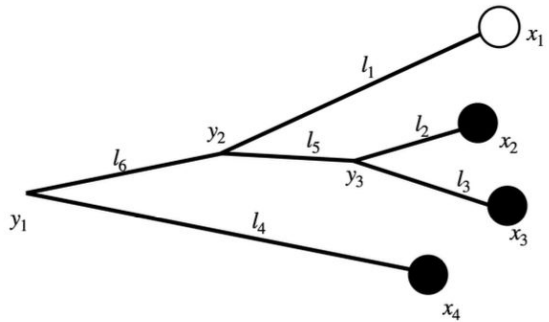
Interpretation for phylogeography

The likelihood we found was the likelihood that the tip states ended up where they did, given the Markov model (rate matrix Q) and the tree:

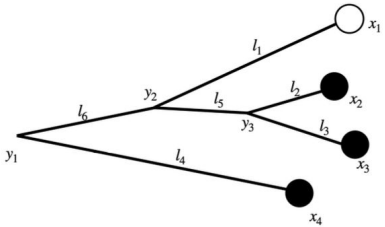
$$P(\mathbf{x} \mid \mathcal{T}) = \sum_{\mathbf{y}} \pi(y_1)p(x_4|y_1)p(y_2|y_1)p(x_1|y_2)p(y_3|y_2)p(x_2|y_3)p(x_3|y_3)$$

This is a **marginalization of a joint distribution**. It has the form

$$p(\mathbf{x} \mid \mathcal{T}) = \sum_{\mathbf{y}} p(\mathbf{x}, \mathbf{y} \mid \mathcal{T}).$$



The joint distribution



We seek the set of internal node states $\mathbf{y}$ that gives the *biggest* contribution to the sum, ie the largest $p(\mathbf{x}, \mathbf{y} \mid \mathcal{T})$.

Hard problem! Solution: while performing Felsenstein's pruning, *record* the values of $L_{\text{node}}(\text{node state})$ that are computed.

Start by setting $\hat{y}_1 = \text{argmax}_z \pi(z) L_{y_1}(z)$ as the most likely root state.

Then, given $\hat{y}_1$, set $\hat{y}_2 = \text{argmax}_{y_2} p\hat{y}_1, z(l_6) L_{y_2}(z)$. This looks complicated. But it's just "set y_2 to the state that gives the highest contribution".

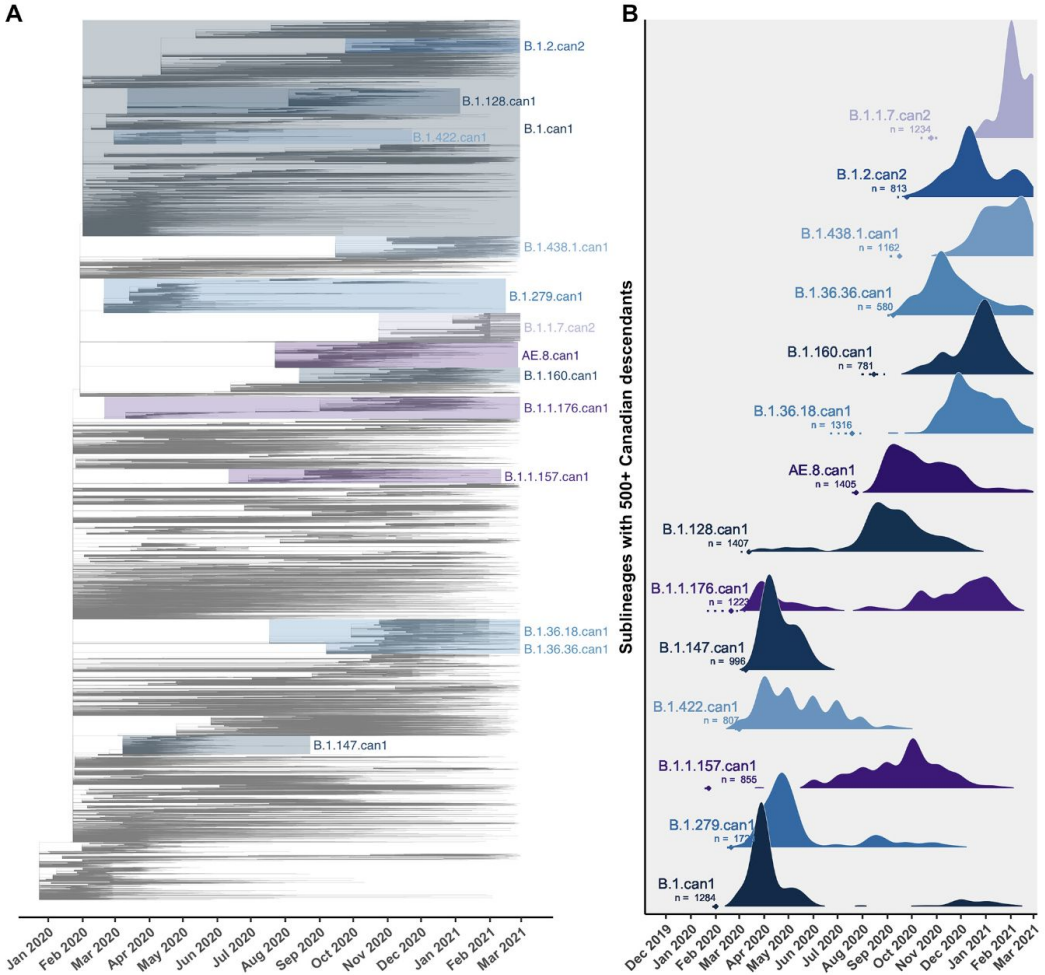
Then, continue down the tree, setting each node's state to the one that makes the highest contribution to $\sum_{\mathbf{y}} p(\mathbf{x}, \mathbf{y} \mid \mathcal{T})$.

COVID-19 application

50,000 sequences including most available Canadian sequences (2021)

Left: phylogeny, coloured by COVID lineage

Right: times of nodes of lineages with > **500 Canadian descendants**. This inference was done with phylogeography (ace).



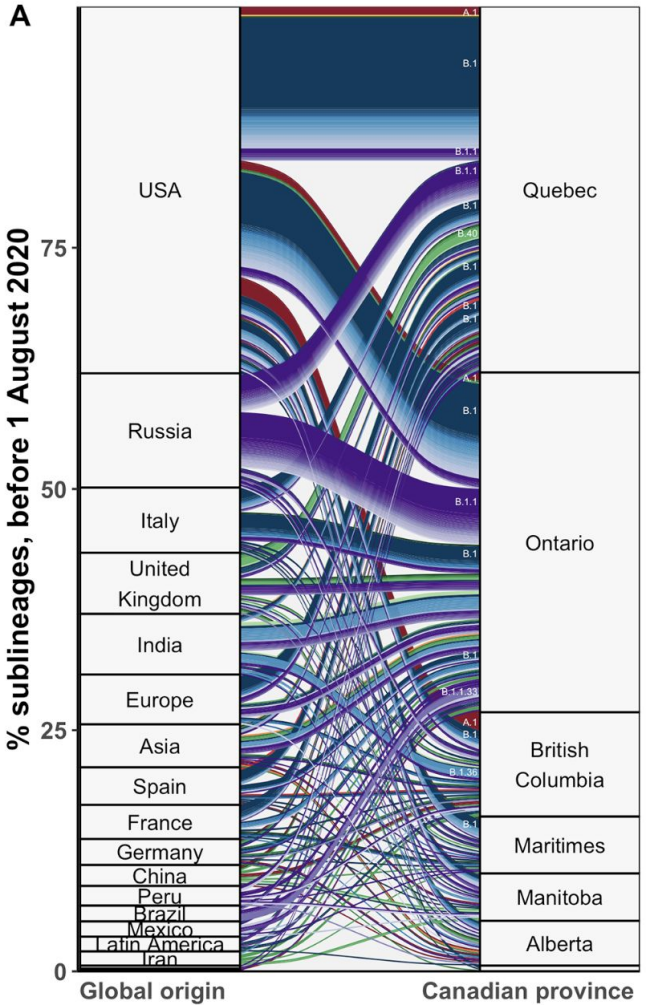
Genomic epidemiology of the first two waves of SARS-CoV-2 i

Angela McLaughlin [1,2,✉], Vincent Montoya [1], Rachel L Miller [1,2], Gideon J Mordecai [3]; Canadian Genomics Network (CanCOGen) Consortium[‡], Michael Worobey [4], Art FY Poon [5], Jeffrey B Joy

COVID-19 movements

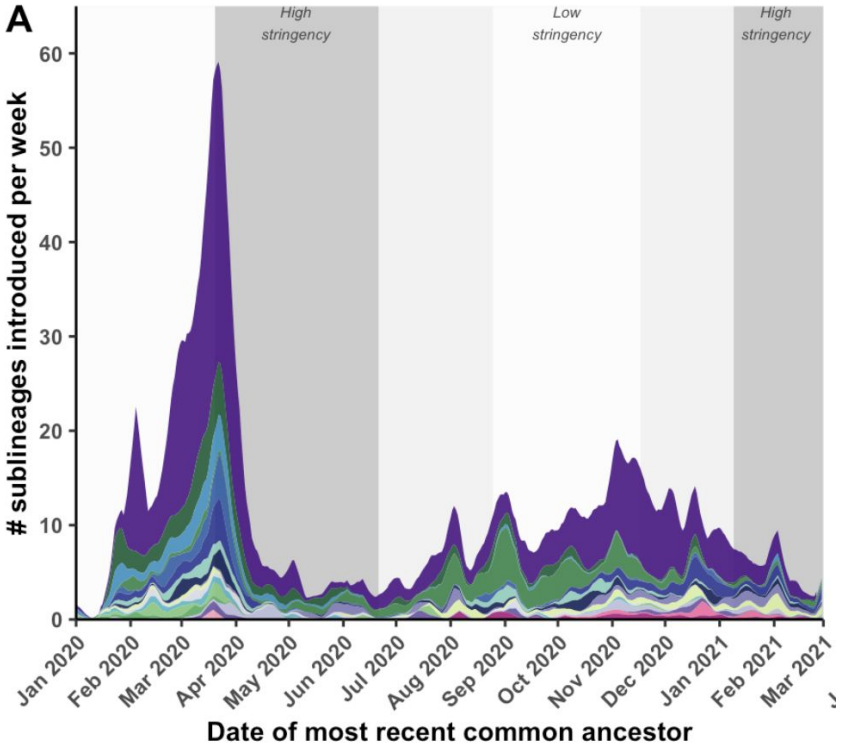
These “alluvial plots” illustrate the inferred viral movements.

They are formed by counting edges with a parent node in one state (eg Russia) and the child node in another (eg Ontario, a province of Canada).



Number of introductions (that are ancestral to the sample)

The authors also asked how many edges were there in the tree with ancestral states **y**, at a given time, with the parent node outside Canada and the child node in Canada?



Bayesian phylogeography: Lemey et al 2009

The tree is not known with certainty.

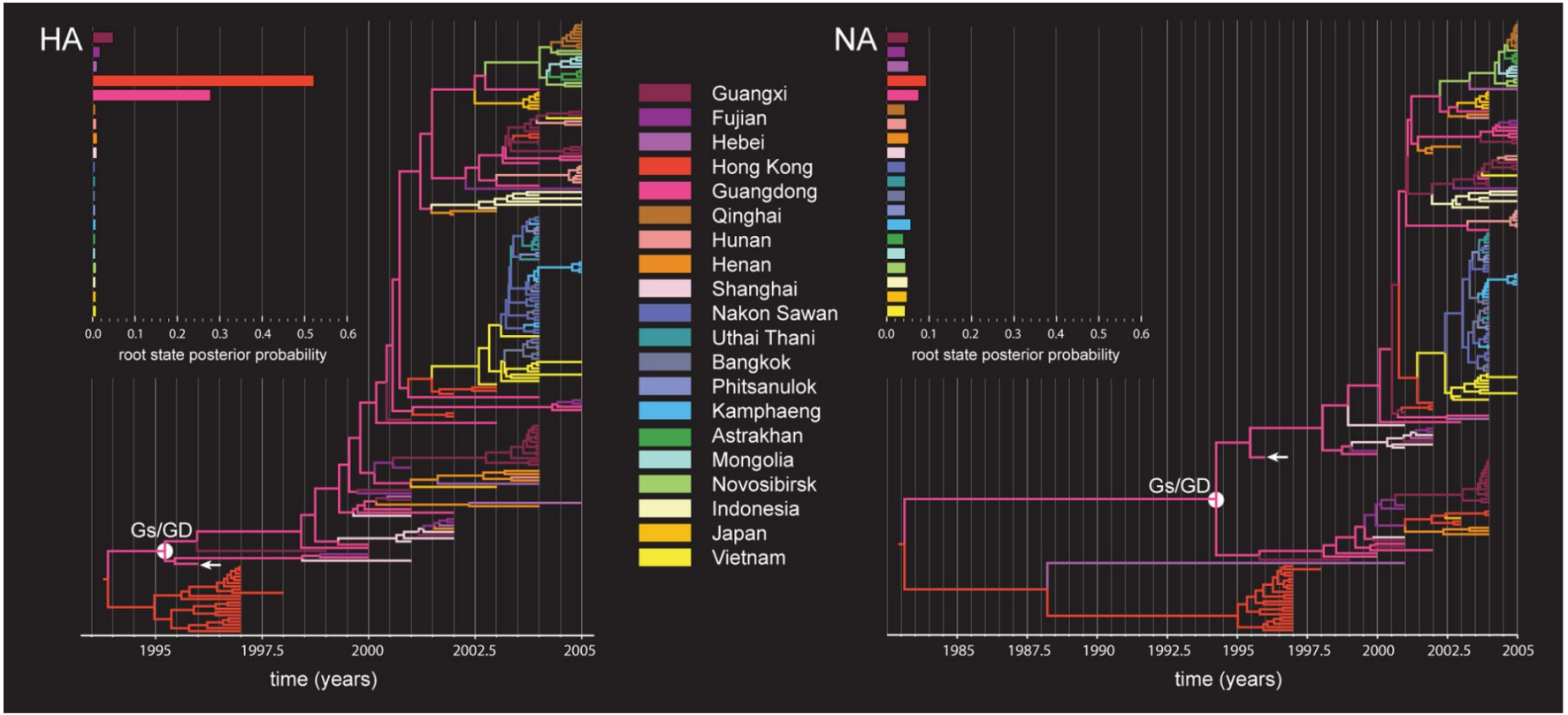
The Bayesian approach samples a posterior:

$$P(T, Q, \mu \mid \mathbf{x}, D) \propto P(\mathbf{x} \mid T, Q)\, P(D \mid T, \mu)\, P(T)\, P(Q)\, P(\mu)$$

where $\mathbf{x}$ are the observed tip locations, D is the sequence data, T is the tree, Q is the location rate matrix, and μ is the clock rate (and substitution model parameters).

The two likelihoods $P(\mathbf{x} \mid T, Q)$ (which is what we just derived) and $P(D \mid T, \mu)$ are independent given T: the location and sequence CTMCs run on the same tree but are otherwise separate. (Note, this is an assumption!)

Phylogeography of 192 A-H5N1 strains, 20 locations



Lemey et al 2009

A problem with discrete trait phylogeography

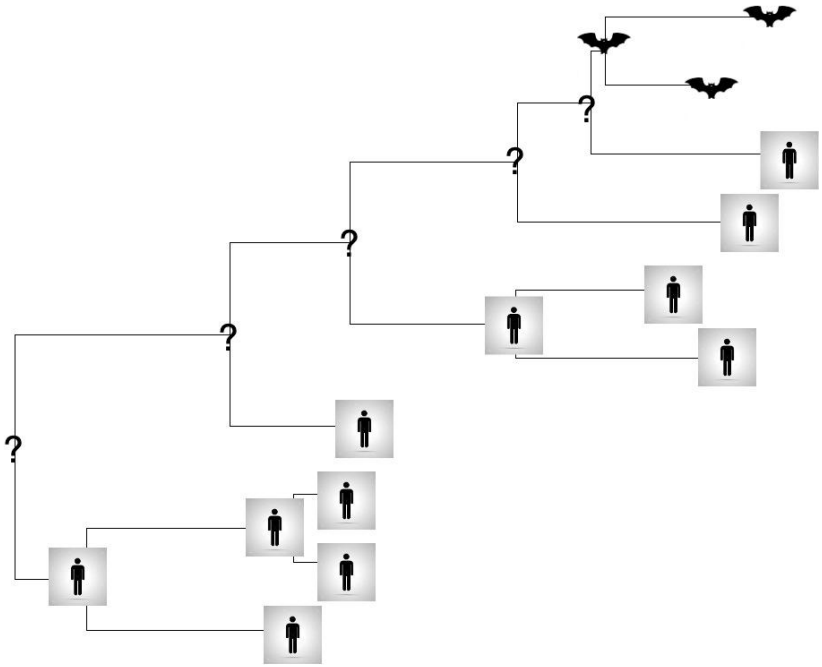
Sampling differences in different states can greatly bias the ancestral state reconstructions.

Note that there is no term related to how likely sampling is, in our computations.

Conceptual example: Ebola virus (right)

The internal nodes: what state are they in? Should they be in bats or humans?

A Markov model like ace will likely place all but the last one in humans. (Why?)



Aim: large-scale phylogeography with adjustment for known sampling differences

- Suppose the UK samples and sequences more of its SARS-CoV-2 infections than the USA.
- Or: the USA sequences more influenza viruses than many countries.
- Large-scale viral phylogeography analyses (10,000s of taxa) and tools like nextstrain use downsampling to adjust for this. (eg the COVID-19 paper we just saw)
- But this is time-consuming and discards a lot of data.
- Our approach: Sampling-aware ancestral state inference (SAAS), *Nature Communications,* 2026 (Song, McPherson, Colijn)

State-dependent Speciation and Extinction (SSE) models

These models began with a paper by Maddison, Midford and Otto, Systematic Biology, 2007 (BiSSE: Binary SSE) and were not used for phylogeography. Their goal: the branching in the tree depends on the state.

In our application, we specify rates at which a lineage in state $i \in \mathcal{S}$ (e.g. location, host species) does the following things:

Event	Rate
Speciates (transmits), giving daughter states j, k	$\Lambda_{i,j,k}$
Dies (recovers/clears)	μ_i
Is sampled	ψ_i
Transitions from state i to j	q_{ij}

Two main quantities in the SSE model

We compute two key quantities (generalizations of the the birth–death–sampling quantities g_e, p_0):

Let $D_{e,i}(\tau) = P(\mathcal{T}_e(\tau) \mid Y_e(\tau) = i, \theta)$ be the probability that edge e in state i at (backward) time τ gives rise to the *observed* subtree below it. This is like g_e in our birth-death model.

Let $E_i(\tau) = P(\text{no observed descendants} \mid \text{lineage in state } i \text{ at time } \tau)$. This is like $p_0(\tau)$.

Differential equation for the Ds

$D_{e,i}(\tau)$ satisfies (like the ODE for g_e):

$$\frac{d}{d\tau} D_{e,i} = -\left(\sum_{j,k} \Lambda_{i,j,k} + \mu_i + \psi_i + \sum_{j \neq i} q_{ij}\right) D_{e,i} + \sum_{j,k} 2\Lambda_{i,j,k}\, D_{e,j}\, E_k + \sum_{j \neq i} q_{ij}\, D_{e,j}$$

Initial conditions at a node $\tau_{e,0}$:

$$D_{e,i}(\tau_{e,0}) = \begin{cases} \psi_i & \text{(sampled tip)} \quad \longleftrightarrow \quad g_e(s) = \psi \\ \displaystyle\sum_{j,k} \Lambda_{i,j,k}\, D_{e_1,j}(\tau_{e,0})\, D_{e_2,k}(\tau_{e,0}) & \text{(speciation node)} \quad \longleftrightarrow \quad g_e(s_1) = \lambda\, g_{e_1} g_{e_2} \end{cases}$$

Differential equation for the Es

$E_i(\tau)$ satisfies (like the ODE for p_0):

$$\frac{d}{d\tau}E_i = \mu_i - \left(\sum_{j,k}\Lambda_{i,j,k} + \mu_i + \psi_i + \sum_{j\neq i} q_{ij}\right)E_i + \sum_{j,k} 2\Lambda_{i,j,k}\,E_j\,E_k + \sum_{j\neq i} q_{ij}\,E_j,$$

with $E_i(0) = 1$ (at the present, a lineage has probability 1 of leaving no future observed descendants).

With just one state and no Q (no state transitions) these equations reduce to the previous birth-death-sampling model equations

What came before our edge? Condition on the "complement tree"

$D_{e,i}$ and E_i have information *below* edge e: how likely is the tree that comes after edge e in state i?

To infer ancestral states we also need information *above* edge e (the "ancestral complement" $\mathcal{T}_e^C$). Define (forward in time $t = T - \tau$):

$$\widetilde{D}_{e,i}(t) = \Pr\big(Y_e(T-t) = i \mid \mathcal{T}_e^C(T-t), \theta\big)$$

For *anagenetic* (state-transition only) changes:

$$\frac{d}{dt}\widetilde{D}_{e,i}(t) = -\sum_{j \neq i} q_{ij}\, \widetilde{D}_{e,i}(t) + \sum_{j \neq i} q_{ji}\, \widetilde{D}_{e,j}(t)$$

These are initialized at each node from the parent edge and sister subtree (see paper for full expression).

Combining the forward and backward information

Ancestral state probability at edge e:

Probability edge is in state i given Tc

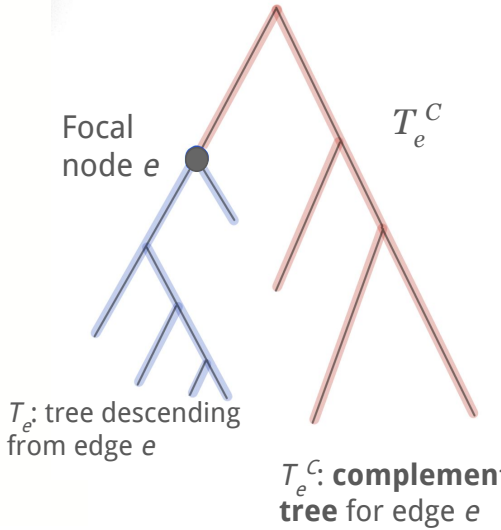
Likelihood of tree below edge *e* given that the state of *e* is *i*

$$A_{e,i}(\tau) = \frac{\widetilde{D}_{e,i}(T-\tau)\, D_{e,i}(\tau)}{\sum_j \widetilde{D}_{e,j}(T-\tau)\, D_{e,j}(\tau)}$$

The denominator normalises: $\sum_i A_{e,i}(\tau) = 1$.

Key innovation (SAASI): ψ_i differs by state.

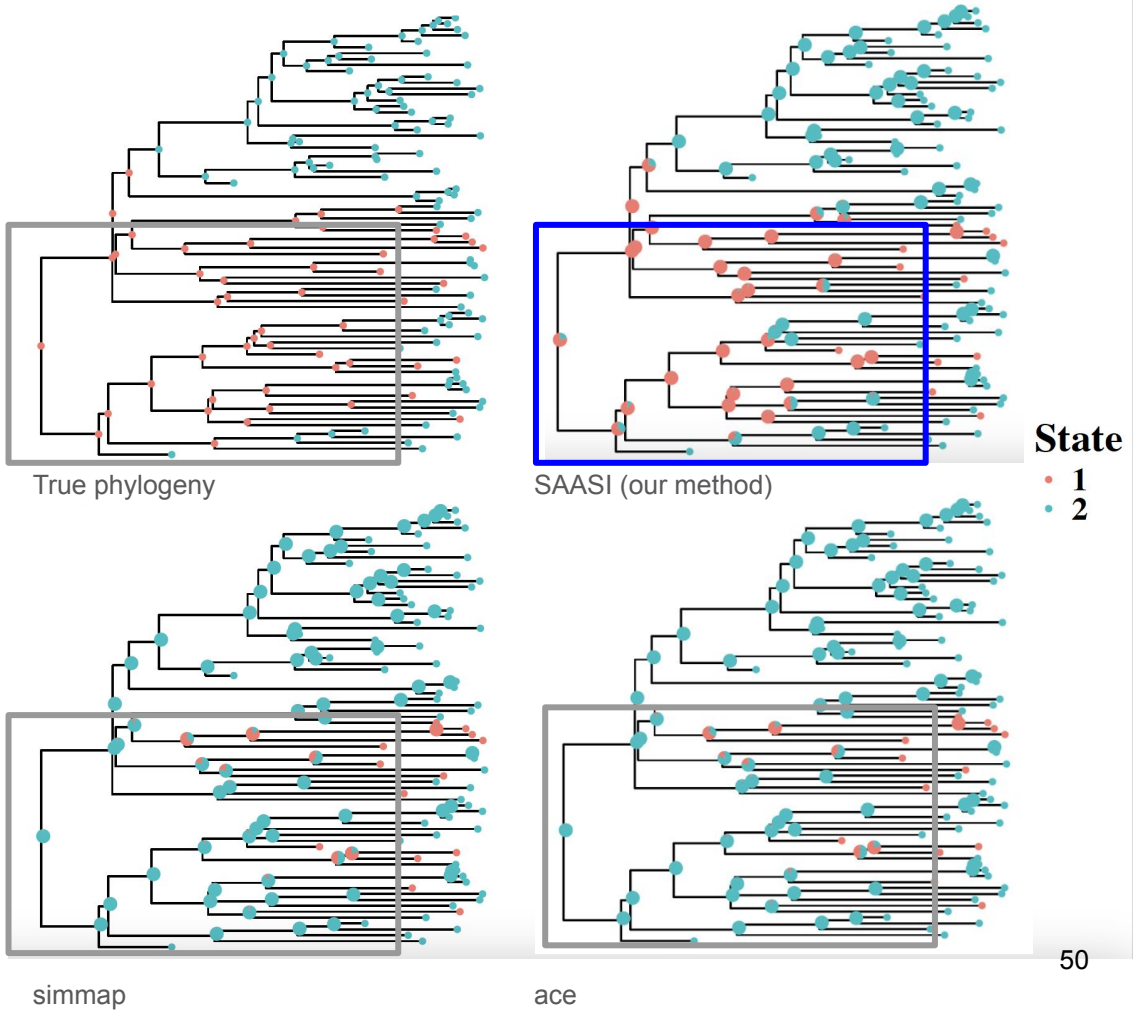
SAASI corrects for this via E_i and $D_{e,i}$.



Example with two states

- State 1 (red) **50** times less sampling than State 2 (cyan)
- Accuracy: fraction of nodes with correct inferred location
- ace & simmap: affected by sampling; 58% accuracy
- SAAI infers ancestral states more accurately (88% in this example).

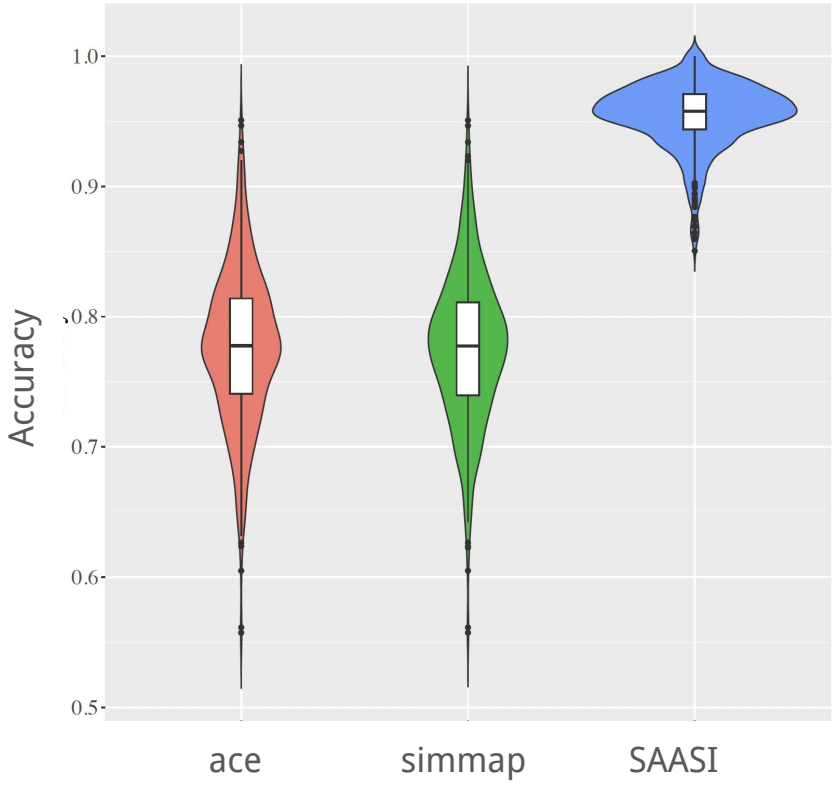
State 2 is over-represented in the tree: ace and simmap wrongly place too many nodes in state 2.



Accuracy over 500 simulated trees

- Simulation: 500 trees. 2 states. Known states at all internal nodes.
- Assume State 1 has 100 times less sampling than State 2.
- Compare SAASI to two standard methods (ace, simmap)
- Accuracy is measured as the overall accuracy:

  fraction of internal nodes whose internal node state is assigned correctly (max probability state)



H5N1 HA demonstration

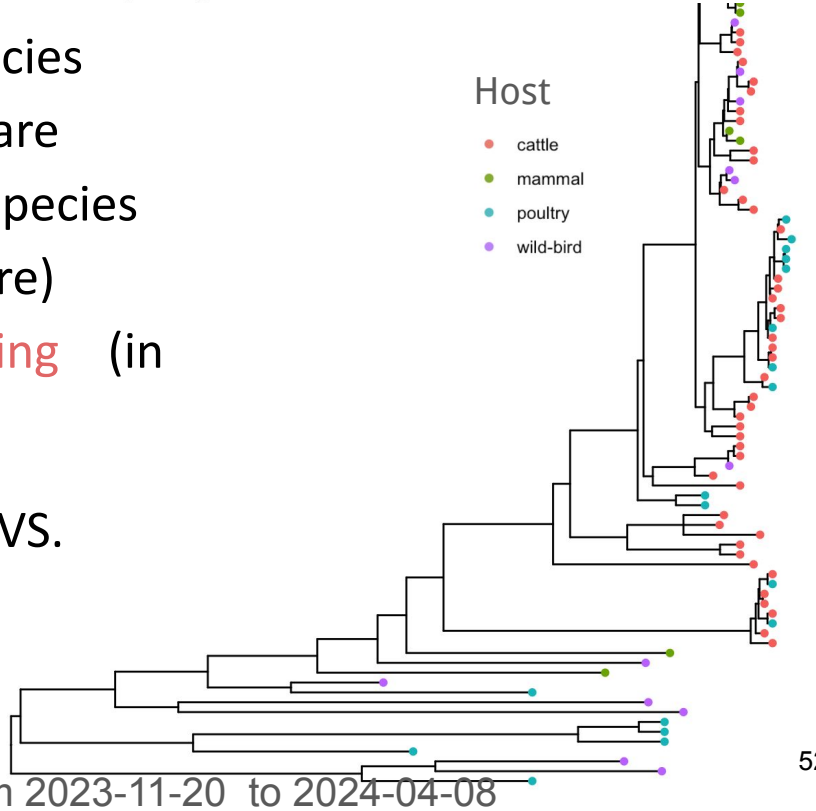
Emergence and interstate spread of highly pathogenic avian influenza A(H5N1) in dairy cattle

Thao-Quyen Nguyen, Carl Hutter, Alexey Markin, Megan Thomas, Kristina Lantz, Mary Lea Killian, Garrett M. Janzen, Sriram Vijendran, Sanket Wagle, Blake Inderski, Drew R. Magstadt, Ganwu Li, Diego G. Diel, Elisha Anna Frye, Kiril M. Dimitrov, Amy K. Swinford, Alexis C. Thompson, Kevin R. Snevik, David L. Suarez, Erica Spackman, Steven M. Lakin, Sara C. Ahola, Kammy R. Johnson, Amy L. Baker, Suelee Robbe-Austerman, Mia Kim Torchetti, Tavis K. Anderson

doi: https://doi.org/10.1101/2024.05.01.591751

- 104 taxa, HA segment, multiple host species
- We suspect that infections in wild birds are undersampled compared to other host species (more accessible, eg domestic, agriculture)
- We want to examine the effect of sampling (in wild birds in particular).

Data: Nguyen *et al* Fig. S2. Used BEAST + BSSVS.



https://github.com//andersen-lab//avian-influenza

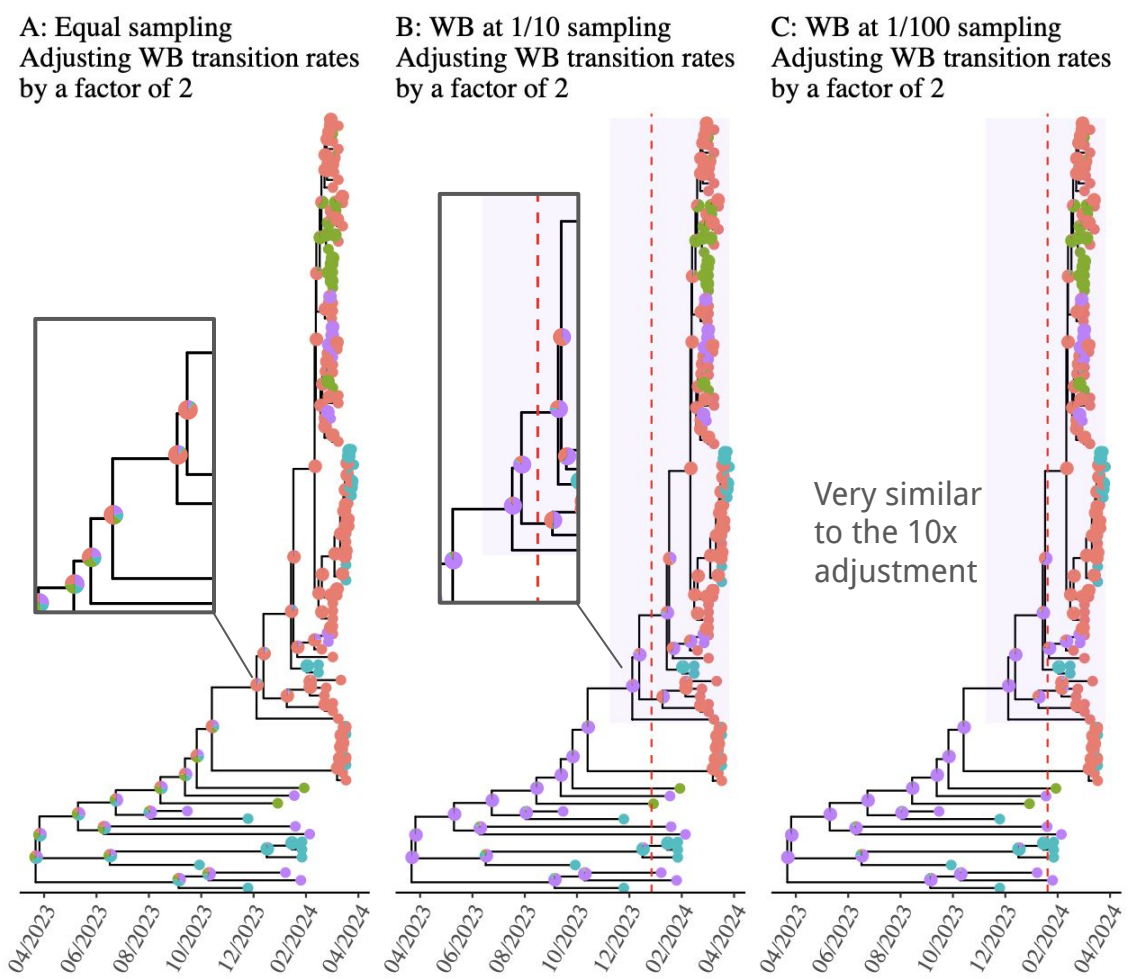
Effect of sampling in wild birds

No sampling adjustment: same result as ace, Nguyen et al:

- one introduction to cattle
- Dec 2023-Jan 2024.

With sampling adjustment:

- Multiple introductions
- late Jan-early Feb 2024.



Method inputs are a challenge

The **bad news**: node state probabilities are defined using differential equations that use state-dependent branching, extinction and sampling rates, and inter-state transition rates. These must be estimated.

For now this is done separately,
but in principle it can be done in the same model

$$\frac{d}{d\tau}D_{e,i}(\tau) = -\left(\sum_{j,k}\lambda_{i,j,k} + \mu_i + \psi_i + \sum_{j\neq i} q_{i,j}\right) D_{e,i}(\tau) + \sum_{j,k} 2\lambda_{i,j,k} D_{e,j}(\tau) E_k(\tau) + \sum_{j\neq i} q_{i,j} D_{e,j}(\tau).$$

The **good news**: results are quite robust to mis-specification of these parameters. We use ace for the transition rates, and use simulation tests to adjust ace's rates.

Method implementation: saasi R package

Inputs:

- A phylogenetic tree; tip states
- Parameter estimates (birth rate, death rate, transition rate matrix)
- Sampling rates (if not known, SAASI works as a sensitivity analysis)

Output:

- The phylogenetic tree with estimated node states (vector of probabilities)

Practicalities

- Time is linear in the number of tips. With 2 states, **100,000 tips**: **1 hour**.
- R package: https://github.com/MAGPIE-SFU/saasi (in development)